04820KA001 Mono Slitter 903_V12_1110_2

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1.PLC

1-8.POUs

1-8-1.Programs

1-8-1-1.TaglMonoTRM

1-8-1-1-9.Tension_Manager

0

```
// ====================== TENSION MANAGER ======================
// 1. Tension Helper: recommended tension from paper GSM, roll width, paper type
// 2. T-one remote set point (AO Ch1 -> isolator -> T-one pins 20/19, F.37 = 1):
//    helper tension x core boost x taper on the rewinder diameter, rate-limited,
//    1..10 V = TensMin_kg..TensMax_kg (T-one F.38 / F.39)
// TensAOEnable = FALSE -> AO Ch1 = fixed 10 V (old Fix10Vdc, pots supply)

// ---- 1. Paper type: KRAFT / BLEACHED selects the factor (N/mm width per 100 g/m2)

IF (TensFactorBleached < REAL#0.1) OR (TensFactorBleached > REAL#2.0) THEN
    TensFactorBleached := REAL#0.54;
END_IF;

IF (TensFactorKraft < REAL#0.1) OR (TensFactorKraft > REAL#2.0) THEN
    TensFactorKraft := REAL#0.80;
END_IF;

IF PaperKraft THEN
    TensFactor := TensFactorKraft;
ELSE
    TensFactor := TensFactorBleached;
END_IF;

// ---- 2. Helper inputs (limits)

IF PaperGSM < REAL#10 THEN
    PaperGSM := REAL#10;
ELSIF PaperGSM > REAL#500 THEN
    PaperGSM := REAL#500;
END_IF;

IF WebWidth_mm < REAL#100 THEN
    WebWidth_mm := REAL#100;
ELSIF WebWidth_mm > REAL#3000 THEN
    WebWidth_mm := REAL#3000;
END_IF;

IF (TensUnit < 0) OR (TensUnit > 2) THEN
    TensUnit := 0;
END_IF;

// ---- 4. Recommended tension

TensIdeal_N := PaperGSM / REAL#100 * WebWidth_mm * TensFactor;
TensIdeal_Disp := TensIdeal_N / REAL#9.81;          // kg

TensShowKg := (TensUnit = 0);
TensShowLb := (TensUnit = 1);
```

```
49  TensShown := (TensUnit - 2);
50
51  // ---- 5. Setup limits (maintenance popup)
52
53  IF TensTaper < REAL#0.0 THEN
54      TensTaper := REAL#0.0;
55  ELSIF TensTaper > REAL#60.0 THEN
56      TensTaper := REAL#60.0;
57  END_IF;
58
59  IF TensCoreBoost < REAL#0.0 THEN
60      TensCoreBoost := REAL#0.0;
61  ELSIF TensCoreBoost > REAL#50.0 THEN
62      TensCoreBoost := REAL#50.0;
63  END_IF;
64
65  IF TensRate_kg_s < REAL#1.0 THEN
66      TensRate_kg_s := REAL#1.0;
67  ELSIF TensRate_kg_s > REAL#20.0 THEN
68      TensRate_kg_s := REAL#20.0;
69  END_IF;
70
71  IF TensMin_kg < REAL#0.0 THEN
72      TensMin_kg := REAL#0.0;
73  END_IF;
74
75  IF TensMax_kg < TensMin_kg + REAL#5.0 THEN
76      TensMax_kg := TensMin_kg + REAL#5.0;
77  END_IF;
78
79  IF TensTrim_kg < REAL#-10.0 THEN
80      TensTrim_kg := REAL#-10.0;
81  ELSIF TensTrim_kg > REAL#10.0 THEN
82      TensTrim_kg := REAL#10.0;
83  END_IF;
84
85  IF TensTrim_kg < REAL#-10.0 THEN
86      TensTrim_kg := REAL#-10.0;
87  ELSIF TensTrim_kg > REAL#10.0 THEN
88      TensTrim_kg := REAL#10.0;
89  END_IF;
90
91  IF TensJogBoost < REAL#0.0 THEN
92      TensJogBoost := REAL#0.0;
93  ELSIF TensJogBoost > REAL#100.0 THEN
94      TensJogBoost := REAL#100.0;
95  END_IF;
96
97  // trim back to 0 when the paper changes
98
99  IF (PaperGSM <> TensGSMMem) OR (WebWidth_mm <> TensWidthMem) OR
    (PaperKraft <> TensKraftMem) THEN
100     TensTrim_kg := REAL#0.0;
101 END_IF;
102
103 TensGSMMem := PaperGSM;
104 TensWidthMem := WebWidth_mm;
105 TensKraftMem := PaperKraft;
106
```

```
107   // ---- 6. Rewinder diameter fraction 0 (core) .. 1 (max Ø)
108
109   IF (RwMaxDiam > RwCoreDiameter + REAL#50.0) AND NOT RwSensFault THEN
110       TensDiaFrac := (RwDiam - RwCoreDiameter) / (RwMaxDiam - RwCoreDiameter);
111   ELSE
112       TensDiaFrac := REAL#0.0;                    // DT35 fault: no taper
113   END_IF;
114
115   IF TensDiaFrac < REAL#0.0 THEN
116       TensDiaFrac := REAL#0.0;
117   ELSIF TensDiaFrac > REAL#1.0 THEN
118       TensDiaFrac := REAL#1.0;
119   END_IF;
120
121   // ---- 7. Profile preview (HMI): core / mid / max
122
123   TensAtCore := TensIdeal_N / REAL#9.81 * (REAL#1.0 + TensCoreBoost / REAL#100.0);
124   TensAtMid  := TensAtCore * (REAL#1.0 - TensTaper / REAL#100.0 * REAL#0.5);
125   TensAtMax  := TensAtCore * (REAL#1.0 - TensTaper / REAL#100.0);
126
127   // ---- 7b. Threading / knife adjustment preset: JOG raises the tension, START
      resumes normal
128
129   IF JogRun THEN
130       TensJogLatch := TRUE;
131   END_IF;
132
133   IF StartPB_OK AND NOT StartBlocked THEN          // START accepted -> normal
      tension
134       TensJogLatch := FALSE;
135   END_IF;
136
137   // ---- 8. Target: profile (taper) + trim, or manual; x JOG boost while threading
138
139   IF TensAuto THEN
140       TensTarget := TensAtCore * (REAL#1.0 - TensTaper / REAL#100.0 * TensDiaFrac) +
      TensTrim_kg;
141   ELSE
142       TensTarget := TensManual_kg;
143   END_IF;
144
145   IF TensJogLatch THEN
146       TensTarget := TensTarget * (REAL#1.0 + TensJogBoost / REAL#100.0);
147   END_IF;
148
149   IF TensTarget < TensMin_kg THEN
150       TensTarget := TensMin_kg;
151   ELSIF TensTarget > TensMax_kg THEN
152       TensTarget := TensMax_kg;
153   END_IF;
154
155   // ---- 9. Rate limit (4 ms scan): smooth change of GSM / width / type / mode
156
157   TensStep := TensRate_kg_s * REAL#0.004;
158
159   IF TensSet_kg < TensTarget - TensStep THEN
160       TensSet_kg := TensSet_kg + TensStep;
161   ELSIF TensSet_kg > TensTarget + TensStep THEN
162       TensSet_kg := TensSet_kg - TensStep;
```

```
163 ELSE
164     TensSet_kg := TensTarget;
165 END_IF;
166
167 // ---- 10. Analog output Ch1
168
169 IF TensAOEnable THEN
170     // 1..10 V (400..4000) = TensMin_kg..TensMax_kg ; 0 V = wire broken (below
    range)
171     TensAOReal := REAL#400.0 + REAL#3600.0 * (TensSet_kg - TensMin_kg) /
    (TensMax_kg - TensMin_kg);
172
173     IF TensAOReal < REAL#400.0 THEN
174         TensAOReal := REAL#400.0;
175     ELSIF TensAOReal > REAL#4000.0 THEN
176         TensAOReal := REAL#4000.0;
177     END_IF;
178
179     AO_TOne_SetPoint := REAL_TO_INT(TensAOReal);
180 ELSE
181     AO_TOne_SetPoint := INT#4000;              // fixed 10 V (pots supply, as Fix10Vdc)
182 END_IF;
```

1-9.Data

1-9-2.Global Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Network Publish	Comment
VAR_GLOBAL							
Pot1_Work_Velocity	INT		BuiltInIO://opt/#0/Ch1_Analog_Input_Value	False	False	Publish Only	Potentiometer 1 value - Work speed -
InvRef	INT		BuiltInIO://opt/#0/Ch2_Analog_Output_Value	False	False	Publish Only	Inverter speed reference
AO_TOne_SetPoint	INT		BuiltInIO://opt/#0/Ch1_Analog_Output_Value	False	False	Publish Only	Analog O to T-one Tension SetPoint [Kg]
VelMax	REAL	800		True	False	Publish Only	Maximum speed
VelSet	REAL			False	False	Publish Only	Set speed
RwSlowDecelK	REAL	0.15		True	False	Publish Only	Learned slow-down decel per mm of rewinder diameter (mm/s2 per mm)
RwMaxSlowFallback	REAL	1285.0		True	False	Publish Only	Rewinder max: slow-down diameter while K not learned (mm)
Language	DINT			True	False	Publish Only	Language
TensJogLatch	bool	False		False	False	Publish Only	threading tension active (HMI lamp)
TensTrim_kg	REAL	0.0		True	False	Publish Only	Operator trim on the AUTO tension (kg), ±10
TensJogBoost	REAL	30.0		True	False	Publish Only	Threading preset: % extra while JOG / until START (0–100)
RwThickMin_mm	REAL	0.02		True	False	Publish Only	Learning: min plausible paper thickness (mm)
PulseRate	UDINT		IOBus://unit#3/Ch1 Pulse Rate	False	False	Publish Only	Pulses per 100 ms
RwThickMax_mm	REAL	2.0		True	False	Publish Only	Learning: max plausible paper thickness (mm)
RwDecelMin_mm	REAL	20.0		True	False	Publish Only	Learning: min plausible stop decel (

							mm/s2)
RwDecel Max_mm	REAL	5000.0		True	False	Publish Only	Learning: max plausible stop decel (mm/s2)
RwLearnF ailMax	INT	12		True	False	Publish Only	Learning: failed windows before alarm (12 = 2 min)
RwStopFa ilMax	INT	3		True	False	Publish Only	Learning: failed stops before alarm
RwThick_ mm	REAL			True	False	Publish Only	Learned paper thickness (mm)
OverSpeed	BOOL			False	False	Publish Only	Speed too high
WorkSpd_ Max	REAL	500.0		True	False	Publish Only	Work speed pot full scale (m/min), 50..VelMax
WorkSpd_ Max_Set	REAL	400.0		True	False	Publish Only	HMI request: work pot full scale (m/min), applied only with motor stopped
PaperKraft	BOOL	False		True	False	Publish Only	Paper type: TRUE = kraft, FALSE = bleached (tension factor)
TensFactor Bleached	REAL	0.54		True	False	Publish Only	Tension factor bleached paper (N/mm width per 100 g/m2)
TensFactor Kraft	REAL	0.80		True	False	Publish Only	Tension factor kraft paper (N/mm width per 100 g/m2), tune on machine
TensAuto	BOOL	False		True	False	Publish Only	T-one set point from the PLC profile (FALSE = TensManual_kg)
TensAOEn able	BOOL	False		True	False	Publish Only	AO Ch1 = T-one set point (FALSE = fixed 10 V as old Fix10Vdc)
TensManu al_kg	REAL	40.0		True	False	Publish Only	T-one manual set point (kg) when TensAuto is OFF
TensTaper	REAL	30.0		True	False	Publish Only	Tension taper core -> max diameter (%), 0..60
TensCoreB oost	REAL	0.0		True	False	Publish Only	Extra tension at the core (%), 0..50
TensRate_ kg_s	REAL	5.0		True	False	Publish Only	T-one set point max change (kg/s), 1..20
TensMin_	REAL	10.0		True	False	Publish Only	T-one set point at 1 V

kg							(= T-one F.38)
TensMax_kg	REAL	100.0		True	False	Publish Only	T-one set point at 10 V (= T-one F.39)
TensSet_kg	REAL	0.0		False	False	Publish Only	Live T-one set point sent on AO Ch1 (kg)
TensAtCore	REAL	0.0		False	False	Publish Only	Profile preview: tension at the core (kg)
TensAtMid	REAL	0.0		False	False	Publish Only	Profile preview: tension at mid diameter (kg)
TensAtMax	REAL	0.0		False	False	Publish Only	Profile preview: tension at max diameter (kg)
UwRawMin	INT	10		True	False	Publish Only	Unwinder sonar raw min (below = sensor fault, alarm 17)
UwRawMax	INT	3990		True	False	Publish Only	Unwinder sonar raw max (above = sensor fault / no echo, alarm 17)
TensCheckEnable	BOOL	False		True	False	Publish Only	Tension check ON (HMI switch); OFF = start as before
TensWaitSpd	REAL	10.0		True	False	Publish Only	Speed held after START until tension established (m/min)
TensionOK	BOOL	False		False	False	Publish Only	Tension established (debounced T-one Alarm 0)
TensWait	BOOL	False		False	False	Publish Only	Waiting for tension (speed held at TensWaitSpd, HMI lamp)
SonarAsk	BOOL	False		False	False	Publish Only	V10 HMI popup: sonar is OFF, turn it on? (HMI NO resets it)
UwSlow_at_D_Req	BOOL	True		True	False	Publish Only	V7c: unwinder diameter slow-down follows the speed (FALSE = fixed UwSlowDown_Diam)
UwSlowDiamAct	REAL	0.0		False	False	Publish Only	V7c: unwinder slows down at this diameter (mm), HMI display
FinalStopDecel	REAL	500.0		True	False	Publish Only	V12: end-of-roll stop ramp, web decel (mm/s2), 0 = drive DEC ramp as before,

							50..2000
FinalStop Ramp	BOOL	False		False	False	Publish Only	V12: end-of-roll stop ramp running (HMI lamp)
RwJogMaxDiam	REAL	400.0		True	False	Publish Only	V7d: jog blocked above this rewinder diameter (mm), 0 = no limit
JogInhibit	BOOL	False		False	False	Publish Only	V7d: jog blocked (rewinder roll too big), HMI lamp
SCurveTime	REAL	0.5		True	False	Publish Only	V7f: S-curve rounding of the AUTO ramps (s), 0 = off, max 2
TOneDecel	BOOL	False		False	False	Publish Only	V7f: line decelerating (for a T-one stop/boost input)
HornStartBeep_s	REAL	1.0		True	False	Publish Only	V9 horn: beep when the slow-down starts (s), 0 = off, max 5
HornBeepLen_s	REAL	0.2		True	False	Publish Only	V9 horn: length of the short beeps during slow-down (s), 0 = off, max 2
HornBeepPeriod	REAL	5.0		True	False	Publish Only	V9 horn: one short beep every ... s during slow-down, 0 = off, max 30
HornStopBeep_s	REAL	3.0		True	False	Publish Only	V9 horn: beep when stopped after a slow-down (s), 0 = off, max 10
HornTest	BOOL	False		False	False	Publish Only	V9 horn: HMI test button (momentary), machine stopped only
RwLockDiam	REAL	300.0		True	False	Publish Only	V9: paper thickness LOCKED at this rewinder diameter (mm), 0 = never lock
RwLockMinGrow_mm	REAL	50.0		True	False	Publish Only	V9: min diameter growth since the start of learning before locking (mm)
RwMatLocked	BOOL	False		True	False	Publish Only	V9: paper thickness locked (HMI lamp), NEW PAPER / GSM YES unlocks
LongThick_mm	REAL	0.0		False	False	Publish Only	V9: thickness from the long baseline (mm

							), HMI display
RwDiamRev	REAL	0.0		False	False	Publish Only	V9: rewinder diameter averaged over one revolution (mm)
RwWave_mm	REAL	0.0		False	False	Publish Only	V9: diameter wave peak-peak in the last revolution (mm), HMI
UwStopLenShow	BOOL	False		False	False	Publish Only	V9 HMI: show unwinder STOP LENGTH (AUTO STOP UNWINDER active)
UwSlowDiamShow	BOOL	False		False	False	Publish Only	V9 HMI: show unwinder SLOW AT diameter (normal operation, sonar ON)
SoftCancelAccel	REAL	100.0		True	False	Publish Only	V9: then ramp up at this web accel (mm/s2), 0 = AccelWeb
RwMatReset	BOOL	False		False	False	Publish Only	HMI NEW PAPER: forget the learned paper thickness RwMat_K (PLC resets it)
FinalSpd_Max_Set	REAL	80.0		True	False	Publish Only	HMI request: final pot full scale (m/min), applied only with motor stopped
FinalSpd_Max	REAL	80.0		True	False	Publish Only	Final speed pot full scale (m/min), 5..WorkSpd_Max
UnderSpeed	BOOL			False	False	Publish Only	Speed too low
RwLearnReset	BOOL	False		False	False	Publish Only	HMI / Watch: clear learned decels RwStopDecelK, RwStopDecel, RwSlowDecelK (PLC resets it)
RwCoreDiameter	REAL	100.0		True	False	Publish Only	Rewinder core outside diameter for TEACH CORE (mm)
RwTeachCore	BOOL	False		False	False	Publish Only	HMI: TEACH CORE request (PLC resets it)
RwTeachState	INT	0		False	False	Publish Only	TEACH CORE: 0 idle 1 teaching 2 done 3 refused
SpdDelta_	REAL	12.0		True	False	Publish Only	Speed difference perc

Percent							entage
Encoder_Pos	DINT		IOBus://unit#3/Ch1 Encoder Present Position	False	False	Publish Only	Encoder counter
UwTeachState	INT	0		False	False	Publish Only	Unwinder TEACH CORE: 0 idle 1 teaching 2 done 3 refused
Encoder_On	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Counter Enable	False	False	Publish Only	Encoder enable
Encoder_Reset	BOOL		IOBus://unit#3/Ch1 Encoder Counter Operation Command/Ch1 Internal Reset Execution	False	False	Publish Only	Encoder reset
HMI_Screen	ARRAY [0..1] OF DINT			False	False	Publish Only	HMI panel Screen
Velocity_M_min	REAL			True	False	Publish Only	Speed in m/min
Current_Metre	REAL			True	False	Publish Only	Current metres
Speed_Display	REAL			True	False	Publish Only	Set speed (displayed)
I_ControlVoltage	BOOL		BuiltInIO://cpu/#0/Input_Bit_00	False	False	Publish Only	In_0 ControlVoltageON
I_Start_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_01	False	False	Publish Only	In_1 Start button
I_Stop_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_02	False	False	Publish Only	In_2 Stop button

I_Motor_Thermal	BOOL		BuiltInIO://cpu/#0/Input_Bit_03	False	False	Publish Only	In_3 Motor thermal switch
I_Prot_By Pass	BOOL		BuiltInIO://cpu/#0/Input_Bit_04	False	False	Publish Only	In_4 Guards-bypassed selector switch - Under Panel
I_Hub_Position	BOOL		BuiltInIO://cpu/#0/Input_Bit_05	False	False	Publish Only	In_5 Hub/chuck in extraction position
I_FrontGuard_Closed	BOOL		BuiltInIO://cpu/#0/Input_Bit_06	False	False	Publish Only	In_6 Front guard closed
I_Jog_PB	BOOL		BuiltInIO://cpu/#0/Input_Bit_07	False	False	Publish Only	In_7 Jog Push button
I_VFD_Alarm	BOOL		BuiltInIO://cpu/#0/Input_Bit_08	False	False	Publish Only	In_8 Inverter in alarm
I_Side_Door_Closed	BOOL		BuiltInIO://cpu/#0/Input_Bit_13	False	False	Publish Only	In_Side Door is Closed
I_OR_Position_Sens	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN01 (Pin 1.1)_6000_01	False	False	Publish Only	OverRider position sens
I_OR_Hoover_Mode	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN02 (Pin 2.1)_6000_02	False	False	Publish Only	OverRider Hoover Mode
I_OR_Auto_OS	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-EC	False	False	Publish Only	OverRider in automatic

			O Input-_IN03 (Pin 1.2) _6000_03				
I_OR_Cont_OS	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN04 (Pin 2.2) _6000_04	False	False	Publish Only	OverRider in continuous
O_MotorStart	BOOL		BuiltInIO://cpu/#0/Output_Bit_00	False	False	Publish Only	Q_0 Motor start
I_FanTemp_Ok	BOOL		ECAT://node#[1,0]/IB IL 24 DI 8/HD-ECO Input-_IN06 (Pin 2.3) _6000_06	False	False	Publish Only	Motor (fan) thermal OK
O_Alarm_Array	BOOL		ECAT://node#[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT01 (Pin 1.1) _7000_01	False	False	Publish Only	Alarm lamp
O_Prot_ByPass_Lamp	BOOL			False	False	Publish Only	Guards bypassed lamp
O_Motor_Lamp	BOOL		BuiltInIO://cpu/#0/Output_Bit_01	False	False	Publish Only	Q_1 Motor ON lamp
O_RwUnload	BOOL		ECAT://node#[1,1]/IB IL 24 DO 8/HD-ECO Output_OUT03 (Pin	False	False	Publish Only	Unloaders enable - opening

			1.2) _7000_03				
O_FrontGuard_ByPass	BOOL		ECAT://node# [1,1]/IB IL 24 DO 8/HD-ECO Output_OUT04 (Pin 2.2) _7000_04	False	False	Publish Only	Front guard bypass
O_Brake	BOOL		BuiltInIO://cpu/#0/Output_Bit_02	False	False	Publish Only	Q_2 Brake command
O_OR_Auto_Lamp	BOOL		BuiltInIO://cpu/#0/Output_Bit_06	False	False	Publish Only	Q_6 OverRider auto/continuous lamp
O_OverRider_Up	BOOL		BuiltInIO://cpu/#0/Output_Bit_09	False	False	Publish Only	Q_9 Raise OverRider ~10mm Over Round Rewinder roll
O_OverRider_Hold	BOOL		BuiltInIO://cpu/#0/Output_Bit_04	False	False	Publish Only	Q_4 OverRider Lock
O_CoolingFan	BOOL		BuiltInIO://cpu/#0/Output_Bit_03	False	False	Publish Only	Q_3 Brake and motor fans
Fan_Timeout	REAL	0.5		True	False	Publish Only	Fans stop timeout [min]
OverRider_TimeOFF	REAL	0.01		True	False	Publish Only	OverRider photocell OFF timeout (panel)
OverRider_TimeON	REAL	0.05		True	False	Publish Only	OverRider photocell ON timeout (panel)
AlarmOn	BOOL			False	False	Publish Only	Alarm active
BuzzerOn	BOOL			False	False	Publish Only	Panel buzzer command
Msg	INT			False	False	Publish Only	Message: Warnings / Alarms
Index_B	INT			False	False	Publish Only	Alarm index
MotStop	BOOL			False	False	Publish Only	Motor stopped
Clock1s	BOOL			False	False	Publish Only	1 s clock
ManMode	BOOL			True	False	Publish Only	Manual mode
Counter_Reset	BOOL			False	False	Publish Only	Counter reset from panel
I_FrontGu	BOOL		BuiltInIO:	False	False	Publish Only	In_9 Front guard open

ard_Open			//cpu/#0/Input_Bit_09				
O_RwLoading_Enable	BOOL		BuiltInIO://cpu/#0/Output_Bit_07	False	False	Publish Only	Q_7 Unloaders Load enable - closing
JogSpd_SP	REAL	10.0		True	False	Publish Only	Speed in manual
O_AccDec2	BOOL		BuiltInIO://cpu/#0/Output_Bit_05	False	False	Publish Only	Q_5 Second acc / dec ramp
AccelTrim	REAL	0.08		True	False	Publish Only	Acceleration coefficient in run
DecelTrim	REAL	0.08		True	False	Publish Only	Deceleration coefficient in run
DecelCoarse	REAL	0.3		True	False	Publish Only	Acceleration coefficient at start (orig. comment; name suggests decel)
AccelCoarse	REAL	0.3		True	False	Publish Only	Acceleration coefficient at start (orig. comment; name suggests accel)
UwStopDiamShow	BOOL	False		False	False	Publish Only	V11 HMI: show unwinder STOP AT diameter (slow-down reached, decel / final speed)
O_AcousticSignal	BOOL		BuiltInIO://cpu/#0/Output_Bit_08	False	False	Publish Only	Q_8 Horn / Acoustic signal
Pot2_Slow_Velocity	INT		BuiltInIO://opt/#0/Ch2_Analog_Input_Value	False	False	Publish Only	Potentiometer 2 value - Final speed -
I_AutoStop_OS	BOOL		BuiltInIO://cpu/#0/Input_Bit_10	False	False	Publish Only	In_10 Stop on metre counter
I_Input_11_Spare_1255	BOOL		BuiltInIO://cpu/#0/Input_Bit_11	False	False	Do not publish	In_11 SPARE (wire 12.55) Under Panel
I_WebBreak_Monitor	BOOL		BuiltInIO://cpu/#0/Input_Bit_	False	False	Publish Only	In_12 Paper break monitoring ON

			12				
I_Vacuum_OK	BOOL		IOBus://unit#3/Ch1 Reset/External Input Status/Ch 1 External Input 0 Status	False	False	Publish Only	Extractor fan thermal OK
I_T_one_TensionOK	BOOL		ECAT://node# [1,0]/IB IL 24 DI 8/HD-ECO Input -_IN05 (Pin 1.3) _6000_05	False	False	Publish Only	T-one Alarm 0 (15-16): tension established (replaces I_SlowSens_PH)
I_WebBreak_PH	BOOL		ECAT://node# [1,0]/IB IL 24 DI 8/HD-ECO Input -_IN07 (Pin 1.4) _6000_07	False	False	Publish Only	Paper break photocell
UwStop_at_M_Val	REAL	5100		True	False	Publish Only	Unwinder Stop at metres
UwSlowMargin_m	REAL	10		True	False	Publish Only	Unwinder Slow-down metres
LowSpd_Display	REAL			True	False	Publish Only	Slow-down speed (displayed)
RwMaxDiam	REAL	1315.0		True	False	Publish Only	Rewinder max diameter protection (mm)
RwMaxStop	BOOL			False	False	Publish Only	Rewinder max diameter STOP (latched)
Encoder_Pulse	INT	1000		True	False	Publish Only	Encoder pulses/Rotation
RwSlowDecelLrn	REAL	0.0		True	False	Publish Only	Learned real slow-down decel (mm/s2) - 0 = not learned
CountReset_Req	BOOL			False	False	Publish Only	Counter reset requested
Psw	INT			False	False	Publish Only	Password Entry
I_RsetAlarm	BOOL		ECAT://node# [1,0]/IB IL	False	False	Publish Only	Alarm reset

			24 DI 8/HD-ECO Input -_IN08 (Pin 2.4) _6000_08				
EncoderWheel_Circumf	REAL	402		True	False	Publish Only	Encoder wheel (Cylinder / Roller) Circumference [mm]
AlarmCodeSV	ARRAY [0..32] OF BOOL			False	False	Publish Only	Panel alarms
SafeGuard_ByPass	BOOL			False	False	Publish Only	Guards bypassed
FrontGuard_Closed	BOOL			False	False	Publish Only	Front guard closed
SoftStop	BOOL	False		False	False	Publish Only	Soft stop in progress (HMI)
SoftStopEnable	BOOL	True		True	False	Publish Only	STOP button = soft stop (False = immediate as before)
SoftStopEndSpd	REAL	8.0		True	False	Publish Only	Soft stop: run command drops below this speed (m/min)
UwDiam_Measure	REAL			True	False	Publish Only	Measured distance
UwCoreDiameter	REAL	100		True	False	Publish Only	Core Outside diameter [mm]
Core_Measure	BOOL	False		True	False	Publish Only	Stored core value
UwCoreDistance	REAL	785.56		True	False	Publish Only	Sonar distance to empty core. [mm]
AI_Diam_Unwinder	INT		IOBus://unit#2/Ch1 Analog Input Value	False	False	Publish Only	Analog value - UnWinder Diameter
Measure_ON	BOOL	True		True	False	Publish Only	Unwinder Diameter measurement enabled
UwDiam_Act	REAL			True	False	Publish Only	Measured roll diameter
UwSlow_at_D_Val	REAL	200		True	False	Publish Only	Unwinder Slow-down diameter [mm]
UwStopDiam_Limit	REAL	110		True	False	Publish Only	Stop diameter
Alarm_Reset	BOOL			False	False	Publish Only	Alarm reset from panel
Psw22	BOOL			False	False	Publish Only	Password 22 for Popu

							p 22
Psw44	BOOL			False	False	Publish Only	Password 44 for Popup 44
I_Roll_OK	BOOL			False	False	Publish Only	Roll OK / Broken Holder Joint
AI_Diam_Rewinder	INT		IOBus://unit#2/Ch2 Analog Input Value	False	False	Publish Only	Analog value - ReWinder Diameter
RwStopEnable	BOOL	False		True	False	Publish Only	Rewinder diameter stop enabled
RwTargetDiam	REAL	300.0		True	False	Publish Only	Rewinder final diameter (mm)
RwUseSlow	BOOL	True		True	False	Publish Only	Slow down before target
RwSlowMargin_m	REAL	10.0		True	False	Publish Only	Rewinder Metres at slow speed before stop
RwSlowDecel	REAL	0.0		True	False	Publish Only	Rewinder Slow-down decel mm/s2 - 0 = auto from DecStart
RwResetHyst	REAL	50.0		True	False	Publish Only	Reset requests below target - this [mm]
RwDist0V	REAL	0.0		True	False	Publish Only	DT35 distance at 0 V (mm)
RwDist10V	REAL	782.0		True	False	Publish Only	DT35 distance at 10 V (mm)
RwRawMin	INT	10		True	False	Publish Only	Raw below = sensor fault
RwRawMax	INT	3900		True	False	Publish Only	Raw above = sensor fault
RwStopOnFault	BOOL	True		False	False	Publish Only	Stop on sensor fault
RwAxisDist	REAL	727.0		True	False	Publish Only	Sensor to shaft centre (mm) - teach
RwFilter_K	REAL	0.2		True	False	Publish Only	Distance filter per 100 ms (0..1)
RwMinPulses	DINT	1000		True	False	Publish Only	Min pulses in window to learn
RwLearn_K	REAL	0.3		True	False	Publish Only	Material factor filter weight
RwMat_K	REAL	0.0		True	False	Publish Only	Learned material factor
RwLearnStop	BOOL	True		True	False	Publish Only	Learn stop decel on every stop
RwMinLearnSpd	REAL	15.0		True	False	Publish Only	Min speed to learn a stop (m/min)
RwStopDe	REAL	0.0		True	False	Publish Only	Learned stop decel (p

cel							ulses/s2)
RwTreact	REAL	0.03		True	False	Publish Only	Stop reaction time (s)
RwTstopDef	REAL	3.0		True	False	Publish Only	Stop time until decel learned (s)
RwTrim	REAL	1.0		True	False	Publish Only	Stop length trim (>1 = earlier)
RwDist	REAL			False	False	Publish Only	Filtered distance (mm)
RwDiam	REAL			False	False	Publish Only	Rewinder diameter (mm)
RwSpeed	REAL			False	False	Publish Only	Line speed from encoder (m/min)
RwRemain_m	REAL			False	False	Publish Only	Metres to target diameter
RwStopLen_m	REAL			False	False	Publish Only	Stopping length now (m)
RwSlowLen_m	REAL			False	False	Publish Only	Rewinder Slow-down length [m]
RwStop_at_D_Req	BOOL			False	False	Publish Only	Rewinder diameter STOP request
RwSlow_at_D_Req	BOOL			False	False	Publish Only	Rewinder diameter SLOW request
RwSensFault	BOOL			False	False	Publish Only	Rewinder DT35 sensor fault
UwStopEnable	BOOL	False		True	False	Publish Only	Unwinder Auto Stop - Enable
SlowAtDisp	REAL			False	False	Publish Only	HMI: slow-down starts at (m or mm, see SlowAtMode)
SlowAtMode	INT			False	False	Publish Only	HMI: 0=none 1=unwinder [m] 2=rewinder [mm]
SlowAtShow	BOOL			False	False	Publish Only	HMI: show slow-down display
UwSlow_at_M_Val	REAL			False	False	Publish Only	Unwinder: slow-down starts at [m]
UwSlow_at_M_Req	BOOL			False	False	Publish Only	Unwinder metre slow-down request (latched)
RwSlow_at_D_Val	REAL			False	False	Publish Only	Rewinder slow-down starts at diameter [mm]
RwStop_at_D_Val	REAL			False	False	Publish Only	Rewinder: stop command at diameter (mm)
RwMaxSlow	BOOL			False	False	Publish Only	Rewinder max diameter SLOW (latched)
UwEmpty	REAL	30.0		True	False	Publish Only	Unwinder Empty Hys

Hyst							teresis
GSM_Learned_OK	BOOL			False	False	Publish Only	Paper [Grams / Square Meter] FALSE until GSM learned
RwMaxSlowAtDiam	REAL	1370.0		False	False	Publish Only	Rewinder max: slow-down starts at diameter (mm)
RwStopDecelK	REAL	0.0		True	False	Publish Only	Learned stop decel per mm of rewinder diameter (mm/s2 per mm)
SoftStopDecel	REAL	500.0		True	False	Publish Only	Soft stop decel (mm/s2), 0 = DecelCoarse
SpdMax_Pending	BOOL	False		False	False	Publish Only	New pot range waiting for the motor to stop
RwAxisDist_Prev	REAL	727.0		True	False	Publish Only	Rewinder: RwAxisDist before the last TEACH (for UNDO)
UwCoreDist_Prev	REAL	785.56		True	False	Publish Only	Unwinder: UwCoreDistance before the last TEACH (for UNDO)
RwTeachUndo	BOOL	False		False	False	Publish Only	HMI: UNDO rewinder TEACH (PLC resets it)
UwTeachUndo	BOOL	False		False	False	Publish Only	HMI: UNDO unwinder TEACH (PLC resets it)
Psw33	BOOL			False	False	Publish Only	Password 33 for Popup 33
RwMatChangePct	REAL	35.0		True	False	Publish Only	Paper change detection: window deviation (%), 2 windows in a row = new paper, 0 = off
RwSlowFallbackMm	REAL	30.0		True	False	Publish Only	Rewinder: slow-down this many mm before the target while the paper thickness is not learned (0 = off)
RwAutoMaxSpd	REAL	500.0		True	False	Publish Only	Rewinder AUTO stop ON: maximum work speed (m/min), 0 = no limit
RwWinLen_m	REAL	60.0		True	False	Publish Only	Paper thickness learning: MAX windo

							w length (m of paper)
RwWinGrow_mm	REAL	5		True	False	Publish Only	Paper thickness learning: window = this diameter growth (mm), 4 m .. RwWinLen_m
GsmAsk	BOOL	False		False	False	Publish Only	HMI popup: new parent roll, did you change the paper? (HMI NO resets it)
GsmAskYes	BOOL	False		False	False	Publish Only	HMI popup YES: forget the learned thickness (PLC resets it)
DecelWeb	REAL	250.0		True	False	Publish Only	V7: AUTO decel ramp, real web mm/s2 (0 = DecelCoarse)
AccelWeb	REAL	200.0		True	False	Publish Only	AUTO accel ramp, real web mm/s2 (0 = AccelCoarse)
RwRatioD	REAL	3366.0		True	False	Publish Only	V7: drive counts per m/min x rewinder diameter (mm), adapted while running
FFEnable	BOOL	True		True	False	Publish Only	V7: feed-forward speed loop ON (FALSE = old loop of rung 16)
SpdHold	BOOL	False		False	False	Publish Only	V9: speed held after START during a soft stop, START again = ramp up (HMI lamp)
SonarAskYes	BOOL	False		False	False	Publish Only	V10 HMI popup YES: sonar ON (PLC resets it)
UwStop_at_D_Val	REAL	0.0		False	False	Publish Only	V11: unwinder stop command diameter (mm) so the web stops at UwDiamStop, HMI "STOP AT"
Clock10ms	BOOL			False	False	Publish Only	10mS clock
Clock100ms	bool			False	False	Publish Only	100mS clock
Sec_to_Min	REAL			False	False	Do not publish	
Psw48	BOOL			False	False	Publish Only	Password 48 for Popup 48
JogRun	BOOL			False	False	Publish Only	V10: JOG really running (button, not a

							normal run, not a soft stop)
GuardStopOK	BOOL			False	False	Publish Only	V11: stopped for real -> guard bypass allowed (Output 13)
SlowDown	BOOL			False	False	Publish Only	Slow down
WarningOn	bool			False	False	Publish Only	HMI Warning Display
AlarmSilenced	BOOL			False	False	Publish Only	
PaperGSM	REAL			True	False	Publish Only	
WebWidth_mm	REAL			True	False	Publish Only	
TensFactor	REAL			True	False	Publish Only	Tension Helper
TensShowKg	BOOL			True	False	Publish Only	Tension Helper
TensShowLb	BOOL			True	False	Publish Only	Tension Helper
TensShowN	BOOL			True	False	Publish Only	Tension Helper
TensIdeal_Disp	REAL			True	False	Publish Only	Tension Helper
TensUnit	INT			True	False	Publish Only	Tension Helper
TensUnitPB	BOOL			False	False	Publish Only	Tension Helper
TensLostWarn	BOOL	False		False	False	Publish Only	Tension lost while running (warning)
UwSensTest	BOOL	False		False	False	Publish Only	Unwinder sonar fault (raw out of range 1 s)
UwSensFault	BOOL			False	False	Publish Only	UnWinder Sensor Fault
Thickness_Fault	BOOL			False	False	Publish Only	Thickness Not Learned !
AutoStopWhy	INT			False	False	Publish Only	
AutoStopDone	BOOL			False	False	Publish Only	
AutoEnRefused	BOOL			False	False	Publish Only	
AutoEnWhy	INT			False	False	Publish Only	
TensWaitHz	REAL	1.0		True	False	Publish Only	
StartBlocked	BOOL	True		False	False	Publish Only	
GuardOpen	BOOL	False		False	False	Publish Only	
SideDoor	BOOL	False		False	False	Publish Only	

	Open							
	TensDiaFrac	REAL	0.0		False	False	Publish Only	Rewinder diameter fraction 0 (core) .. 1 (max)
	Psw40	BOOL			False	False	Publish Only	Password 40 for Popup 40